

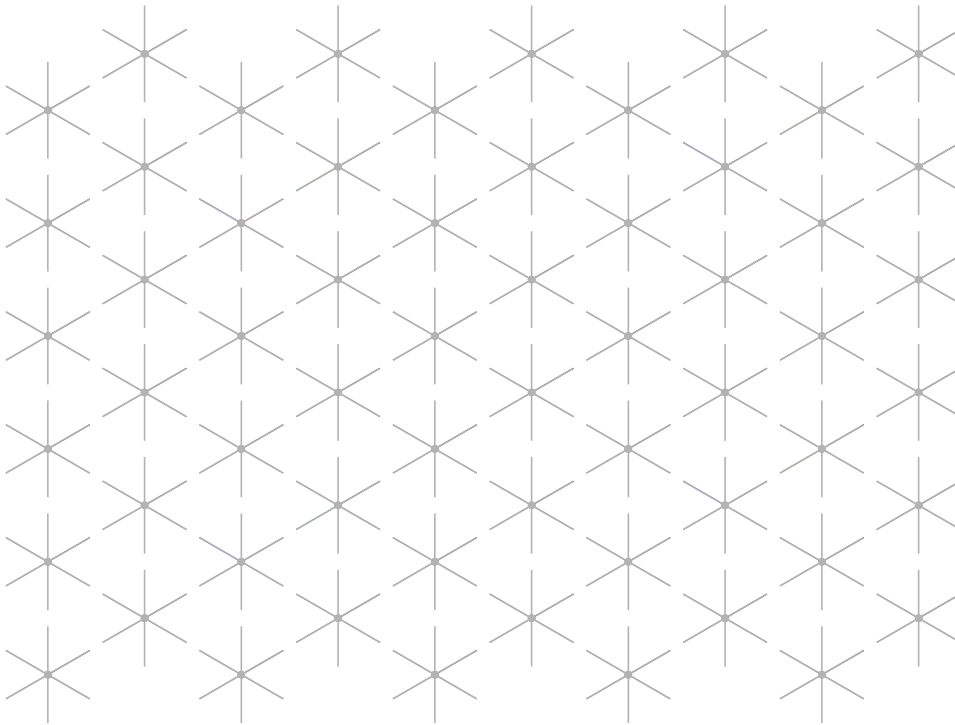
## LAB2 Project

due 26.01.2025, 23:59

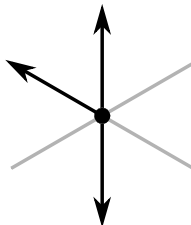
### 1 FHP Model

It has been shown that certain “cellular automata” macroscopically reproduce the dynamics of incompressible Navier Stokes equations. One such cellular automaton is the FHP-1 model. See [2] and [3] for details, but the essentials are:

The 2D world is described by a hexagonal grid, i.e. every node has six neighbors.



Every node can be in one of 64 states: each of the six outgoing directions can be occupied (black arrow in the figure) or not ( $2^6 = 64$ ).

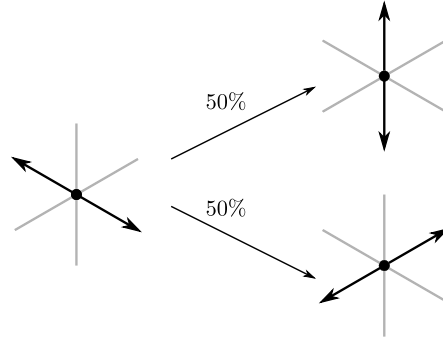


The system is evolved in discrete time steps. Each step consists of two phases. First a collision phase, then a streaming phase.

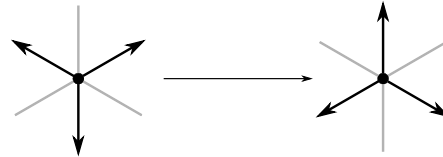
### 1.1 Collision phase

The configuration on each node can be changed (simultaneously) according to the following FHP-1 collision rules

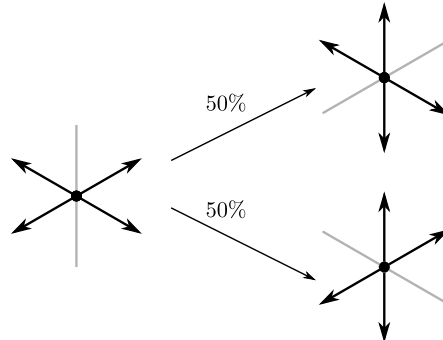
1. A node with exactly two occupied directions, that are back-to-back, is replaced by a node with two different back-to-back occupied directions (50% chance for each)



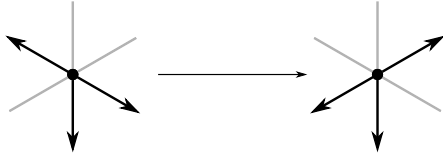
2. A node with three occupied directions, where every second direction is empty, is changed into a node where the other three directions are occupied.



3. A node with four occupied directions, where one back-to-back pair is empty, is changed into a node with four occupied directions, with a different empty back-to-back pair (50% chance for the two different choices)

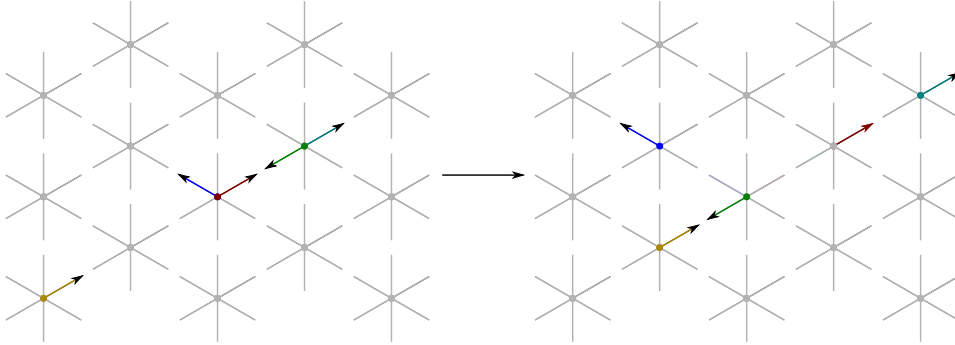


4. A node with three occupied directions, of which two are back-to-back directions and the third occupied direction is the “spectator” is replaced by a node with three occupied directions, with the same spectator direction, but changed back-to-back direction.



## 1.2 Streaming phase

Every occupied direction is moved to the nearest neighbor node in that direction. This happens simultaneously, e.g.:



## 2 Boundary Conditions

Several types of boundary conditions can be employed. The most common choices are listed below.

### 2.1 Periodic

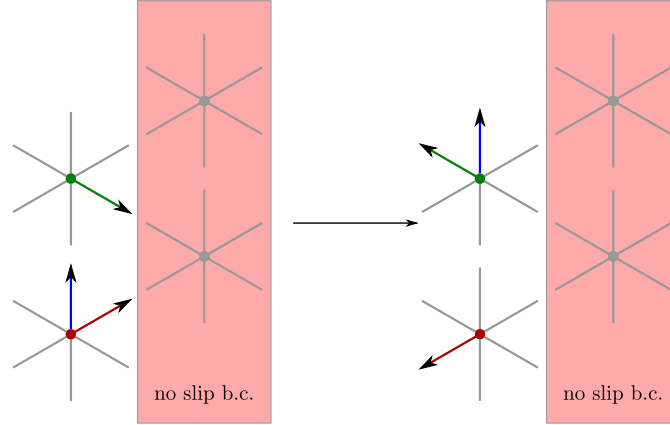
Periodic boundaries are easiest to implement when the number of nodes in each direction is even. One then makes the grid periodic (usually in one of the directions), i.e. the nearest neighbors of the last node in a periodic direction, that would lie outside of the grid, are identified with the matching first nodes of that direction.

### 2.2 Dirichlet

One can declare a node to be a boundary node with Dirichlet boundaries. Such a node is initialized, but then never changed during neither the collision nor the streaming phase. But it influences its nearest neighbors during the streaming phase. That means that there is a constant flow through the boundary at these nodes.

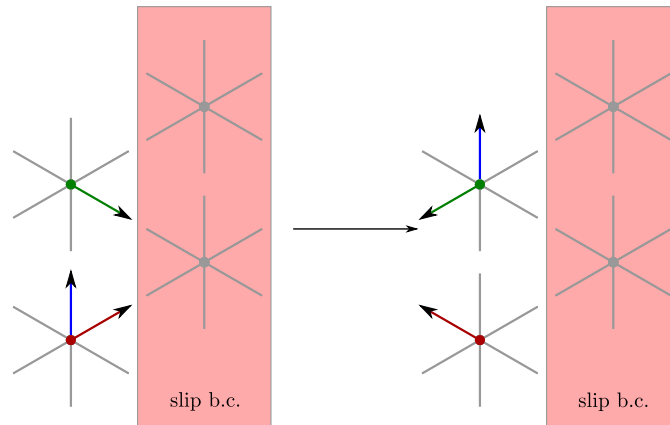
### 2.3 No-slip

Nodes can be declared to be no-slip boundaries. Let us say, node  $A$  is a no-slip boundary and node  $B$  is one of its nearest neighbor nodes. When  $B$  updates its state during the streaming phase, to update one of its “legs” it looks at the state of node  $A$ . In that case, instead the usual streaming, node  $B$  takes the value of its own leg that goes towards  $A$  and puts it on the opposite leg. E.g.



### 2.4 Slip

Very similarly, nodes can be declared to be slip boundaries. For this one also has to declare a direction: there are horizontal, vertical and diagonal slip boundaries. They work similar to no-slip boundaries, but instead of a 180° reflection, only the component normal to the boundary's orientation is flipped. E.g.



### 3 Physical Quantities

If  $\phi_\mu(\vec{x}) \in \{0, 1\}$ ,  $\mu = 0, \dots, 5$  describes the state of a cell at position  $\vec{x}$ , we can assign a mass and a velocity as follows:

$$m(\vec{x}) = \sum_{\mu=0}^5 \phi_\mu(\vec{x}), \quad (1)$$

$$\vec{v}(\vec{x}) = \sum_{\mu=0}^5 \phi_\mu(\vec{x}) \vec{e}_\mu, \quad (2)$$

where  $e_\mu$  are vectors of length 1 in the six directions of the lattice. The mass is measured in multiples of the elementary particle mass. The velocities in  $a$  over time step. Navier stokes equations describe averaged or “macroscopic” quantities. E.g. one can consider  $16 \times 16$  blocks (sometimes called “voxels”) and form averages or sums. The total mass inside of a voxel divided by its area, is the macroscopical density, while the macroscopical velocity is the averaged elemental one.

### 4 Parallelization

Parallelization with MPI is straightforward. The simplest setup is, if the grid is divided up into stripes. If the grid is oriented as in the figures above, the easiest would be to consider vertical stripes, with even number of columns to live on different processes. Before the streaming phase a boundary-exchange needs to happen, i.e. each process has to receive copies of the two columns - one to the left, one to the right - that are neighboring its local grid. No special data-types are necessary if the memory layout is chosen appropriately.

### 5 Task

Write a program that simulates the FHP-1 lattice gas.

- Use C and MPI. Start with a version without parallelization.
- Make it possible to write out configurations, so they can be plotted with MATLAB or some other package. It may be useful to average blocks of the grid, to obtain a “macroscopic” flow, that can be plotted as a quiver plot.
- Demonstrate the correctness of your simulation. Think of simple predictable examples, and try to reproduce results from the literature.
- Once the code is tested, simulate the following setup

- Grid of size  $30 \times 10$  with a variable lattice spacing, e.g. with  $a = 0.1$  that would be  $300 \times 100$  nodes.
- Slip-b.c. along the long direction.
- Dirichlet or periodic b.c. along the short direction. With Dirichlet: they must be such, that there is a flow through the “pipe”, with periodic b.c. the initial condition must be such that there is a flow.
- An obstacle somewhere inside the pipe, e.g. a sphere at position  $(8, 4)$  with radius 1, that has no-slip boundaries.
- In your term paper of about 10 pages
  - Give a short theoretical introduction to the problem
  - Demonstrate that your implementation is correct.
  - Show some interesting results, e.g. the flow around an obstacle at late times.
  - Demonstrate that your parallelization is efficient. Show a strong scaling plot.

## References

- [1] <https://moodle.uni-wuppertal.de/course/view.php?id=40210>
- [2] “Lattice-Gas Cellular Automata and Lattice Boltzmann Models”, D.A.Wolf-Gladrow, Springer
- [3] “Cellular automaton fluids 1: Basic theory”, S.Wolfram, Journal of statistical Physics Vol. 45